

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:17-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

17-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-19(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

For More Material Join: @JEEAdvanced_2024

PHYSICS

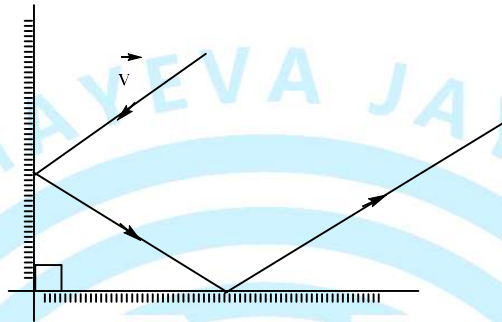
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A particle collides twice, as shown, on two fixed perpendicular surfaces, coming with a speed v . The coefficient of restitution is e for each collision. Neglect gravity. The speed of the particle after the two collisions is



- 1) v 2) ev 3) e^2v 4) $2ev$
2. In Searle's apparatus, used for study of extension of wires verses load, we have two wires. During the experiment, we study the extension in one wire. The use of second wire is ____
- 1) To support the device/ apparatus because it is heavy and may break the single wire.
- 2) To compensate for the change in length caused by changes in temperature of atmosphere during experimentation.
- 3) To keep the apparatus in level so that extension is measured accurately.
- 4) To prevent unnecessary twisting of the instrument during experimentation.
3. **Assertion:** Electric lines of force never cross each other.
Reason: Electric fields at a point superimpose to give one resultant electric field
- 1) Both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- 2) Both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- 3) Assertion is correct Reason is incorrect
- 4) Both Assertion and Reason are incorrect.

4. The rms value of current for an a.c. source producing current

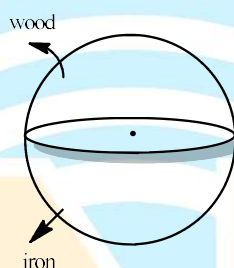
$$I = 2 \sin(100\pi t) + 2 \cos\left(100\pi t + \frac{\pi}{6}\right) \text{ is (in ampere)}$$

- 1) $\sqrt{2}$ 2) $2\sqrt{2+\sqrt{3}}$ 3) 4 4) $2\sqrt{2}$

5. A wooden hemisphere is surmounting on iron made hemisphere (both solids) of radius R. Center of mass of the composite system is at distance $\frac{R}{16}$ from the common center,

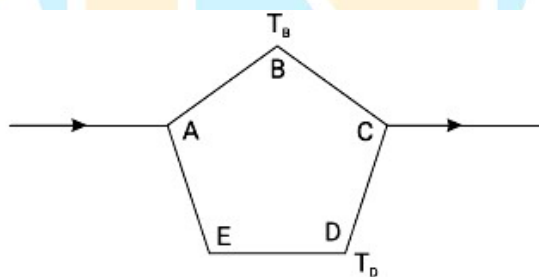
inside the iron – hemisphere, then if the ratio of densities of iron and wood is $\frac{x}{y}$,

then (x + y)



- 1) 16 2) 20 3) 10 4) 12

6. ABCDE is a regular pentagon of uniform wire. The rate of heat entering at A and leaving at C are equal. T_B and T_D are temperatures of B and D. Find the temperature T_C ?



- 1) $\frac{3T_B + 2T_D}{5}$ 2) $3T_D - 2T_B$
 3) $3T_D + 2T_B$ 4) Cannot be determined

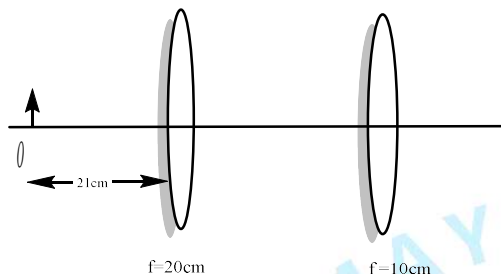
7. An aluminium wire has cross sectional area equal to $8.25 \times 10^{-3} \text{ cm}^2$, and the number density of electrons in aluminium is $9.25 \times 10^{28} \text{ per m}^3$. If current through the wire is 1.5 ampere then drift speed of the electrons (in 10^{-4} m/s) is

- 1) 0.76 2) 1.22 3) 1.68 4) 2.44

8. The work done (in joule) in changing the current of 5 ampere to 8 ampere, by the external source, in a coil of inductance 2 henry is

1) Zero 2) 89 3) 78 4) 39

9.



An object 'O' of size 2 cm is placed at 21 cm from a convex lens of focal length 20 cm. Another convex lens of focal length 10 cm is placed beyond the first lens, at distance 20 cm coaxially. The size of the final image is nearly

1) 2 cm 2) 1 cm 3) 0.5 cm 4) 0.25 cm

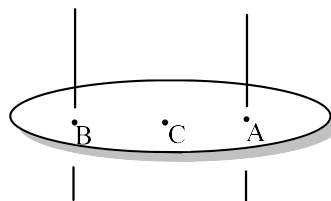
10. The angle of polarisation for a medium is 60° . What will be critical angle for this?

1) $\sin^{-1} \sqrt{3}$ 2) $\tan^{-1} \sqrt{3}$ 3) $\cos^{-1} \sqrt{3}$ 4) $\sin^{-1} \frac{1}{\sqrt{3}}$

11. A proton enters into a room horizontally and is observed to be moving with constant velocity. There may be electrostatic field (E) and magnetic field (B) in that gravity free room. Choose the wrong statement.

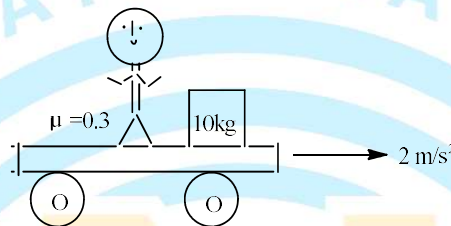
1) $E = 0, B = 0$ 2) $E \neq 0, B = 0$ 3) $E = 0, B \neq 0$ 4) $E \neq 0, B \neq 0$

12. A and B are two points on a plane laminar rigid object, which are collinear with the center of mass C. Given that mass = 20kg; AC = 1.5 cm ; AB = 6.5 cm, and moment of inertia (about the axis through A is 15.9545 kg m^2 . Then the moment of inertia about the axis through B, is (in kg m^2)

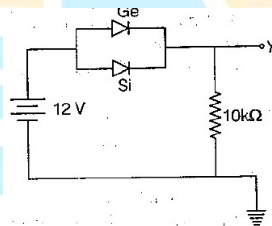


1) 15.5 2) 16 3) 20 4) 25

13. In the experiment for the determination of the speed of sound in air using the resonance column method, the length of the air column that resonates in the fundamental mode, with a tuning fork is 0.1 m. when this length is changed to 0.35 m, the same tuning fork resonates with the first overtone. Calculate the end correction
- 1) 0.012m 2) 0.025m 3) 0.05m 4) 0.024m
14. A 10 kg block kept over the rough surface of a moving trolley is being observed by a person who is also travelling with the trolley. The force of friction between the block and trolley surface with reference to the observer. (Take $g = 10 \text{ m/s}^2$). Answer in newton.



- 1) 30 2) 20 3) 10 4) 0
15. Two junction diodes, one of germanium (Ge) and other of silicon (Si) are connected as shown in the figure, to battery of e.m.f 12 volt and a load resistance $10 \times 10^3 \Omega$. The germanium diode conducts at 0.3 volts and silicon diode at 0.7 volt respectively. The potential difference across the terminals of the load resistance is ____ (volt)



- 1) 15.0 2) 12.0 3) 12.3 4) 11.7
16. A vertical mass-spring system executes simple harmonic oscillations with a period of 2 s. A quantity of this system which exhibits periodic variation with a period of 1 sec is:
- 1) velocity
2) potential energy
3) phase difference between acceleration and displacement
4) Sum of kinetic energy and potential energy

17. In the nuclear reaction ${}_1H^2 + {}_1H^2 \rightarrow {}_2He^3 + {}_0n^1$, If the mass of the deuterium atom = 2.014741 amu, mass of ${}_2He^3$ atom = 3.016977 amu and mass of neutron = 1.008987 amu, then the Q (in MeV) value of the reaction is nearly: (for 1 amu mass energy released is 931 MeV)
- 1) 3.77 2) 4.27 3) 3.27 4) 2.77
18. When radiation of wavelength λ is incident on a metallic surface, the stopping potential is 4.8 volts. If the same surface is illuminated with radiation of double the wavelength, then the stopping potential becomes 1.6 volts. Then the threshold wavelength for the surface is
- 1) 2λ 2) 4λ 3) 6λ 4) 8λ
19. A block of mass $2M$ is attached to a string of length $L = 5$ m. The block is released when the string is horizontal, it picks up a particle of $2M$ kept at rest at the lowest point. The maximum height (in m) attained by the combined mass is (Give answer up to two places of decimal.)
- 1) 0.25 2) 0.50 3) 0.75 4) 1.25
20. The electric field intensity produced by the radiations coming from 100 w bulb at a 3 m distance is E . The electric field intensity produced by the radiations coming from 50 w bulb at the same distance is $(x)^{\frac{-1}{y}} E$. Find $(x + y)$.
- 1) 6 2) 2 3) 4 4) 7

SECTION-II

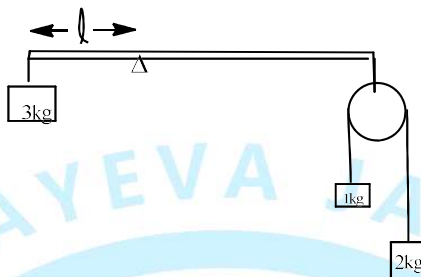
(NUMERICAL VALUE ANSWER TYPE)

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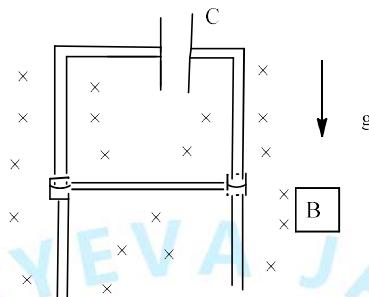
21. The curie temperature of a ferromagnetic material is 300 kelvin. The ratio of magnetic susceptibility of the material (in paramagnetic region) at temperatures 500 kelvin and 400 kelvin is $\left(\frac{x}{y}\right)$. Find $(x+y)$. (where x and y are smallest possible integers)

22. A meter scale of negligible mass is pivoted at a distance (ℓ) from the left end (as shown) such that it remains horizontal. Take $g = 10 \text{ m/s}^2$ and pulley to be mass less and frictionless. Threads are massless. If $\ell = \left(\frac{x}{y}\right)$ meters, then ($y - x$) is _____ (where x and y are smallest possible integers)

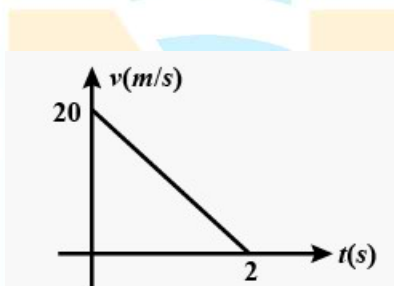


23. At 11°C , the rms speed of a gas molecule is found to be 1000 m/s . Take $R = \frac{25}{3}$ SI units. The molecular mass (in grams), of the gas molecule (in nearest integer) is _____.
24. A spherical ball of density ρ and radius 0.003 m is dropped into a tube containing a viscous fluid. If the viscosity of the fluid $= 1.260 \text{ N.s/m}^2$ and its density is $\rho_L = \rho / 2 = 1260 \text{ kgm}^{-3}$, then find the terminal speed (in cm/s) of the ball. ($g = \text{Acceleration due to gravity} = 10 \text{ ms}^{-2}$)
25. Consider Bohr's atomic theory of single electron system, where the transition of electron from fourth excited state to second excited state of Lithium emits radiation which knocks out the electron from fourth excited state of an atom of an element. The atomic number of the element is _____.
26. A photon possess energy of $2.21 \times 10^{-16} \text{ joule}$. The de-Broglie wavelength associated with the photon is _____ (in $\text{\AA}$)
27. On a hypothetical planet, satellite can only revolve in quantized energy level, i.e., magnitude of energy of a satellite is integer multiple of a fixed energy. Two successive orbits have radii R and $\frac{3R}{2}$. The maximum radius of the satellite in terms of R is nR . Find n .

28. A capacitor ($C = 4 \text{ farad}$) is connected, as shown to a vertical conducting frame, where a horizontal rod glides down without friction in uniform horizontal magnetic field. Take $B = 1 \text{ tesla}$; $\ell = 0.5 \text{ meter}$, mass of rod (m) = 1 kg and $g = 10 \text{ m/s}^2$. The acceleration of the rod while gliding down is $\text{---} \text{ m/s}^2$.



29. Velocity – time graph of a particle of mass 2 kg moving in a straight line is as shown in figure. Work done by all the forces on the particle during first 2 sec has a magnitude of $(50 \times n) \text{ joule}$. Find n .



30. One mole of an ideal gas undergoes a process $P = P_0 \left[1 + \left(\frac{2V_0}{V} \right)^2 \right]^{-1}$, where P_0, V_0 are constants. Change in temperature of the gas when volume is changed from $V = V_0$ to $V = 2V_0$ is given by $\frac{xP_0V_0}{ynR}$. Find the value of $(x + y)$ (where x and y are smallest possible integers)

SECTION – I
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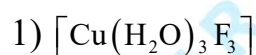
31. Element with which of the following atomic number was named by American society as Rutherfordium while by soviet society it was named as kurchatovium
- 1) 108 2) 104 3) 114 4) 110
32. The reagent suitable for separation of Zn^{2+} and Mn^{2+}
- 1) dil HCl 2) $H_2S + \text{dil HCl}$ 3) Na_2CO_3 4) NaOH
33. Assertion: The mixing of two completely miscible liquids. A and B showing positive deviation from Raoult's law is followed by an absorption of heat
- Reason: The A-B molecular interaction forces are stronger than the A-A or B-B molecular interaction forces.
- 1) Assertion is True, Reason is True; Reason is correct explanation for Assertion
2) Assertion is True, Reason is True; Reason is NOT a correct explanation for Assertion
3) Assertion is True, Reason is False
4) Assertion is False, Reason is True
34. Pick the incorrect statement
- 1) As the atomic number of lanthanide increases, they behave more like aluminium in their chemical behaviour
2) Among the lanthanides samarium has highest melting point.
3) The actinides resemble the lanthanides in having more compounds in +4 oxidation state than in +3 oxidation state.
4) Actinoid contraction is greater from element to element than lanthanoid contraction.

35. Pick up the incorrect statement

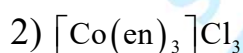
- 1) Cations such as iron (iii), Al (iii) always precipitate as hydroxide with aq ammonia solution
- 2) In presence of NH_3 and CN^- ; $[Co(H_2O)_6]^{2+}$ is unstable
- 3) Co(II) complexes containing NH_3, CN^- ligands are readily oxidised to Co(III) Complexes.
- 4) $[Co(H_2O)_6]^{3+}$ is a high spin complex

36. Set-I (complex Compound)

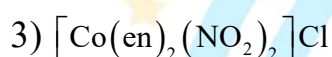
Set-II (Type of Isomerism Shown)



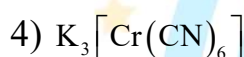
P) Geometrical isomerism



Q) Optical isomerism



R) Linkage isomerism



S) Ionisation isomerism

1) 1 – P, 2 – Q, 3 – R, 4 – S

2) 1 – Q, 2 – P, 3 – S, 4 – R

3) 1 – P, 2 – Q, 3 – S, 4 – R

4) 1 – R, 2 – Q, 3 – S, 4 – P

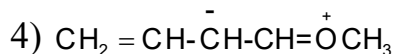
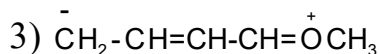
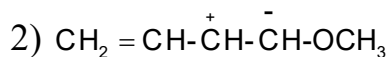
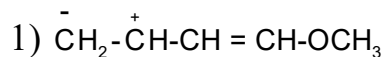
37. Which of the following statements is true about molecular orbital theory?

- 1) Higher the bond order lesser the bond length
- 2) Higher the bond order greater the bond length
- 3) Higher the bond order lesser the bond energy
- 4) Higher the bond order lesser the number of bonds.

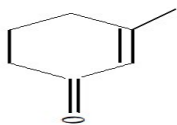
38. Which of the following statement is not true about partition chromatography

- 1) Mobile phase can be a gas
- 2) Separation depends upon equilibrium of solute between a mobile phase and a stationary phase
- 3) paper chromatography is an example of partition chromatography
- 4) stationary phase is a finely divided solid adsorbed

39. Which of the following resonance structure is least stable?



40. $\text{A} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} \text{B} \xrightarrow{\text{OH}^-} \text{Product}$ Correct statement regarding "A" is



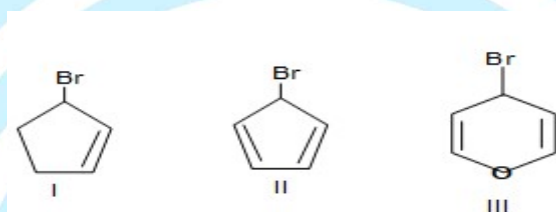
1) it is a alkenyne

2) it is a alkadiene

3) alkadiyne

4) it is a diol

41. Among the given bromides



The order of reactivity in $\text{S}_{\text{N}}1$ reactions is:

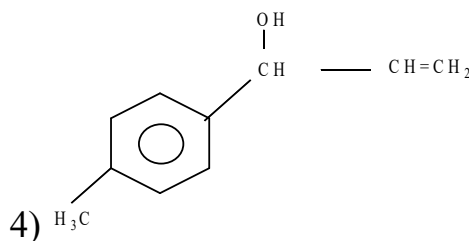
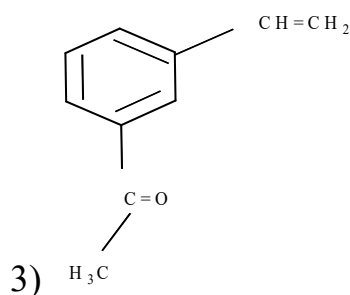
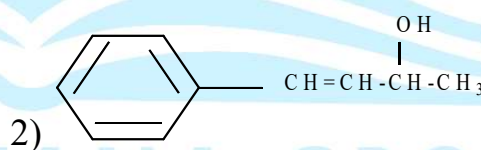
1) $\text{III} > \text{II} > \text{I}$

2) $\text{II} > \text{III} > \text{I}$

3) $\text{III} > \text{I} > \text{II}$

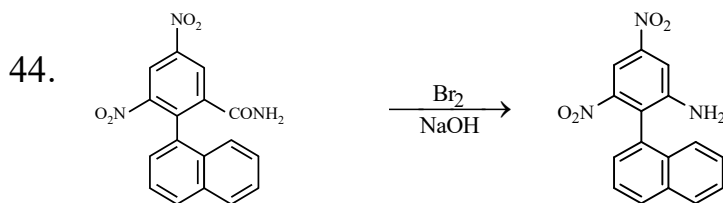
4) $\text{II} > \text{I} > \text{III}$

42. An organic compound containing one oxygen gives red colour with ceric ammonium nitrate solution, decolourises alkaline potassium permanganate solution, respond to iodoform test and exhibit geometrical isomerism organic compound is



1) An ester 2) a carboxylic acid

3) a primary alcohol 4) a 2°- Alcohol.



- 1) The product obtained is optically active and the phenyl group migrates freely during rearrangement.
- 2) The product is an optically active compound and the phenyl group does not become free during migration.
- 3) The product is an optically inactive compound and the phenyl group becomes free during migration.
- 4) The product is an optically inactive compound and the phenyl group does not become free during migration.

1) Adenine 2)Thymine 3) cytosine 4) guanine

1) $3m$ 2) $\sqrt{3}m$ 3) $m/3$ 4) $m/\sqrt{3}$

- 1) for reversible process $\oint \frac{dq_{\text{rev}}}{T} = 0$
- 2) for compression of an ideal gas $|\text{work}_{\text{irreversible}}| \geq |\text{work}_{\text{reversible}}|$
- 3) for irreversible process $\oint ds \neq 0$
- 4) for irreversible process $\oint \frac{dq}{T} > 0$

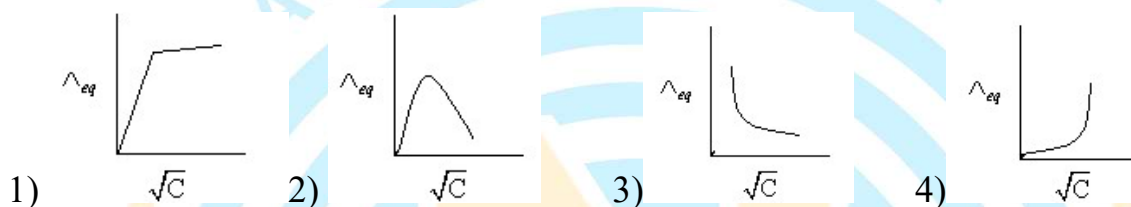
48. When solid mercuric iodide is added to the aqueous solution of potassium iodide then :

- 1) Freezing point is raised 2) Freezing point is lowered
3) Freezing point does not change 4) Boiling point does not change

49. ΔG^0 for the reaction $X + Y \rightleftharpoons C$ is -4.606 Kcal at 1000 K . The equilibrium constant for the reverse mode of the reaction is

- 1) 100 2) 10 3) 1 4) 0.1

50. The variation of equivalent conductance versus $\sqrt{\text{concentration}}$ of a weak electrolyte is correctly given in the plot.



SECTION-II (NUMERICAL VALUE ANSWER TYPE)

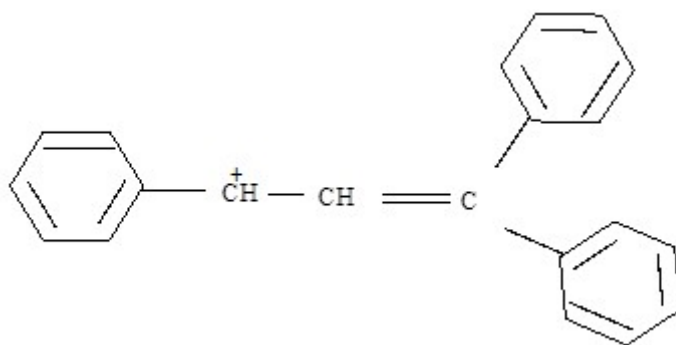
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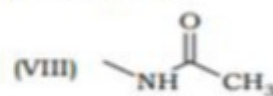
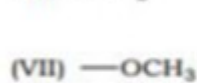
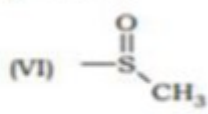
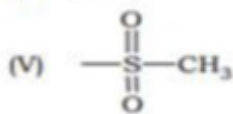
51. when sucrose is subjected to acetylated the molecular weight will be increased by how many units with respective molecular weight of sucrose

52. A diatomic molecule has a dipole moment of 1.2 D . If the bond distance is $1.0 \text{ \AA}$. $\frac{1}{x}$ of an electronic charge "e" exist on each atom, the value of X is

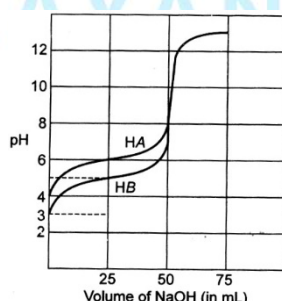
53. Find the total number of position where positive charge can be delocalised by resonance excluding given position of given structure



54. How many of the following shows both (-I) and (+R) effect, if attached to benzene ring



55. $\text{H}-\text{C}\equiv\text{C}-\text{H} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4} (\text{A}) \xrightarrow{\text{dil NaOH}} (\text{B})$; total number of Carbon atom in the product (B)
56. 3-nitro – 5- nitromethyl – cyanobenzene on reduction with LiAlH_4 gives product (A). Atomicity of nitrogen present in the product “A” is
57. $29.2\% \left(\frac{w}{w} \right) \text{HCl}$ stock solution has a density of 1.25 g/ml. The molecular weight of HCl is 36.5gmol. The volume (in ml) of stock solution required to prepare a 200ml solution of 0.4M HCl.
- Topic: atomic structure
58. Find the quantum number “n” corresponding to the excited state of He^+ ion if one transition to the ground state that ion emits two photons in succession with wavelengths 108.5 and 30.4 nm.
59. A gas expands isothermally against a constant pressure of 1atm from 10 L to 20 L. If it also loses an internal energy of 213 J, the heat absorbed for the expansion in joules is (1 lit-atm = 101.3 J)
60. Following is the titration curve of two acids HA and HB (5milli moles each) titrated against strong base NaOH (0.1M)



What is pK_a for HB acid?

MATHEMATICS

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61. The domain of the function $f(x) = \sqrt{2x-x^2} + \frac{1}{\cos^{-1}x} + \frac{1}{\sin^{-1}x}$ is
- 1) (0,2] 2) (0,1] 3) [0,1] 4) (0,1)
62. $R = \{(1, 2), (2, 3), (3, 4)\}$ be a relation on the set of $\{1, 2, 3, 4\}$. Then the least number of elements that must be included in R to get a new relation S where S is an equivalence relation, is
- 1) 5 2) 7 3) 11 4) 13
63. If A is the area between the curve $y = x^6(1-x)^7$ and x -axis then the number of prime factors of A^{-1} is ____
- 1) 3 2) 4 3) 2 4) 5
64. The solution of the differential equation $\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x \cos y}$ is
- 1) $\sin^2 y = x \sin y + \frac{x^2}{2} + C$ 2) $\sin^2 y = x \sin y - \frac{x^2}{2} + C$
- 3) $\sin^2 y = x + \sin y + \frac{x^2}{2} + C$ 4) $\sin^2 y = x - \sin y + \frac{x^2}{2} + C$
65. If the function $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ has a point of maximum at positive values of x then the exhaustive values of ' a ' is
- 1) $\left(-\infty, \frac{29}{7}\right)$ 2) $(-\infty, 7)$
- 3) $(-\infty, -3) \cup \left(3, \frac{29}{7}\right)$ 4) $(3, \infty) \cup (-\infty, -3)$
66. If A and B are two non-zero symmetric matrices, of same order then $AB - BA$ must be
- 1) Symmetric matrix 2) Skew symmetric matrix
- 3) Diagonal matrix 4) Null matrix

67. Let α and β are the roots of $x^2 + 10x - 7 = 0$. If $a_n = \alpha^n + \beta^n$ for $n \geq 1$ then the value of $\frac{a_{12} - 7a_{10}}{2a_{11}}$ is
- 1) 4 2) -5 3) 5 4) -3
68. If $x_1, x_2, x_3, \dots, x_{4001}$ are in an AP such that $\frac{1}{x_1 x_2} + \frac{1}{x_2 x_3} + \dots + \frac{1}{x_{4000} x_{4001}} = 10$ and $x_2 + x_{4000} = 50$ then $|x_1 - x_{4001}| =$
- 1) 20 2) 30 3) 40 4) 100
69. If $f(x)$ is a twice differentiable function in $[0, \pi]$ such that $f(0) = f(\pi) = 0$ then the value of $\int_0^{\pi/2} (f(2x) + f''(2x)) \sin x \cos x dx$ is equal to
- 1) π 2) 2π 3) 3π 4) 0
70. If z_1, z_2 are complex numbers such that $\operatorname{Re}(z_1) = |z_1 - 2|$, $\operatorname{Re}(z_2) = |z_2 - 2|$ and $\operatorname{Arg}(z_1 - z_2) = \pi/3$, then $\operatorname{Im}(z_1 + z_2) =$
- 1) $2/\sqrt{3}$ 2) $4/\sqrt{3}$ 3) $2\sqrt{3}$ 4) $\sqrt{3}$
71. If $y = \frac{\sin^4 x - \cos^4 x + \sin^2 x \cos^2 x}{\sin^4 x + \cos^4 x + \sin^2 x \cos^2 x}$, $x \in \left(0, \frac{\pi}{2}\right)$, then range of y is
- 1) $\left[-\frac{3}{2}, \frac{1}{2}\right]$ 2) $\left[-\frac{5}{3}, 1\right)$ 3) $(-1, 1)$ 4) $[-1, 1]$
72. If $f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \dots + \frac{x^2}{2} + x + 1$ then
- 1) $f'(1) = f'(0)$ 2) $100f'(1) = f'(0)$ 3) $f'(1) = f'(-1)$ 4) $f'(1) = 100f'(0)$
73. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x^3 - 15x^2 + 36x - 48$ is
- 1) is many to one but onto 2) a Bijection
- 3) is an injection but not surjection 4) neither an injection nor a surjection

74. Given that matrix $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$. If $xyz = 2013$ and $8x + 4y + 3z = 2012$ then $A(\text{adj } A)$ is equal to

1) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$ 2) $\begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$ 3) $\begin{bmatrix} 29 & 0 & 0 \\ 0 & 29 & 0 \\ 0 & 0 & 29 \end{bmatrix}$ 4) $\begin{bmatrix} 58 & 0 & 0 \\ 0 & 58 & 0 \\ 0 & 0 & 58 \end{bmatrix}$

75. Exhaustive values of x such that $\cos^{-1}\left(\frac{8x}{1+16x^2}\right) = -\frac{\pi}{2} + 2 \tan^{-1} 4x$

1) $\left(\frac{1}{8}, \infty\right)$ 2) $\left(\frac{1}{16}, \frac{1}{8}\right)$ 3) $\left(\frac{1}{4}, \infty\right)$ 4) $\left(\frac{1}{8}, \frac{1}{4}\right)$

76. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a strictly decreasing function. If $f(2a^2 + a + 1) < f(3a^2 - 4a + 1)$, then the range of $a \in \mathbb{R}$ is

1) $\left(-\infty, \frac{1}{3}\right) \cup (1, \infty)$ 2) $(0, 5)$ 3) $\left(0, \frac{1}{3}\right) \cup (1, 5)$ 4) $[0, 5]$

77. Two dice are thrown simultaneously to get the co-ordinates of a point on xy -plane. Then the probability that this point lies inside or on the region bounded by $|x| + |y| = 3$, is

1) $3/14$ 2) $2/3$ 3) $1/12$ 4) $4/14$

78. The range of random variable X is $\{1, 2, 3, 4, \dots\}$ and the corresponding probabilities are $P(X = K) = \frac{3^{CK}}{K!}$; $K = 1, 2, 3, 4, \dots$, then the value of C is

1) $\log_e 3$ 2) $\log_e 2$ 3) $\log_3(\log_e 2)$ 4) $\log_2(\log_e 3)$

79. In the expansion of $\left(\sqrt[3]{4} + \frac{1}{\sqrt[4]{6}}\right)^{20}$

1) the total no. of rational terms = 4 2) the total no. of irrational terms = 18
3) the middle term is irrational 4) the total no. of irrational terms = 19

80. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x)(3 + \cos x)}{x \tan 4x}$ is equal to

1) $-\frac{1}{4}$ 2) $\frac{1}{2}$ 3) 1 4) 2

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors and $\vec{d}$ be a non-zero vector, which is perpendicular to $\vec{a} + \vec{b} + \vec{c}$. Now, if $\vec{d} = (\sin x)(\vec{a} \times \vec{b}) + (\cos y)(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})$ then minimum value of $x^2 + y^2$ is $k\pi^2/4$ where $k =$
82. Each of the 'n' urns contain 4 white and 6 black balls. The $(n + 1)^{\text{th}}$ urn contains 5 white and 5 black balls. One of the urn is chosen at random and two balls are drawn from it without replacement. Both the balls turn out to be black. If the probability that the $(n + 1)^{\text{th}}$ urn was chosen to draw the balls is $\frac{1}{16}$, then the value of n is
83. The number of points of non-differentiability of the function $f(x) = \begin{cases} \frac{x}{1+|x|}, & |x| \geq 1 \\ \frac{x}{1-|x|}, & |x| < 1 \end{cases}$ is
84. Length of a rectangle increased by 3 units/sec width is decreasing at the rate 2 units/sec, if length is 10 and width is 6 then the area changes by a rate of λ sq.units/sec, then $|\lambda| =$
85. Let $f(x) = ax^2 + bx + c$ with degree less than or equal to 2, $f(0) = 0$, $f(2) = 2$. The minimum value of $\int_0^2 |f'(x)| dx$
86. A curve $g(x) = \int x^{27} (1+x+x^2)^6 (6x^2+5x+4) dx$ is passing through $(0,0)$ then $7.g(1) =$
87. If $A(0,1,2), B(2,-1,3)$ and $C(1,-3,1)$ are the vertices of a triangle, then the distance between its circumcentre and orthocenter is $n/\sqrt{2}$ units, then the value of 'n' is
88. Let T_n be the number of all possible triangles formed by joining vertices of an n-sided regular polygon. If $T_{n+1} - T_n = 10$, then the value of 'n' is
89. If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors, then maximum value of $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ is
90. In an experiment with 15 observations on x, the following results were available:
 $\sum x^2 = 2830, \sum x = 170$. One observation that is 20 was found to be wrong and was replaced by the correct value 30. Then the corrected variance is